Results in Engineering 30 (2026) 110288

Contents lists available at ScienceDirect

Results in Engineering

journal homepage: www.sciencedirect.com/journal/results-in-engineering

Research paper

Data-driven ensemble machine learning for corrosion risk prediction and maintenance optimization in oil and gas pipelines

Adamu Abubakar Sani [a,b,*], Mohamed Mubarak Abdul Wahab [a,c], Nasir Shafiq [a], Kamaluddeen Usman Danyaro [d], Yusuf Aliyu [d]

[a] Department of Civil and Environmental Engineering, Universiti Teknologi PETRONAS, 32610 Seri Iskandar, Perak Darul Ridzuan, Malaysia
[b] Department of Building, Abubakar Tafawa Balewa University, P.M.B. 0248, Bauchi 740272, Nigeria
[c] Center for Urban Resources Sustainability, Institute of Self-Sustainable Building, Universiti Teknologi PETRONAS, 32610 Seri Iskandar, Perak Darul Ridzuan, Malaysia
[d] Department of Computing, Faculty of Science, Management and Computing, Universiti Teknologi PETRONAS, 32610 Seri Iskandar, Perak, Malaysia

ARTICLE INFO

Keywords:
Machine learning
Corrosion risk prediction
Pipeline integrity
Condition-based maintenance
Oil and Gas Pipelines
Predictive Modelling

ABSTRACT

This research focuses on applying machine learning to predict corrosion risks in oil and gas pipelines, with the goal of enhancing efficiency and reducing downtime. Instead of relying on traditional reactive maintenance, the approach adopts a proactive, risk-based strategy that divides pipeline corrosion into three categories: Low, Moderate, and High. Sections with higher corrosion levels are given priority for immediate maintenance, while lower-risk areas are checked less often, allowing resources to be used more effectively. Ensemble machine learning models was developed and evaluated using key performance metrics, including the coefficient of determination (R^2), mean squared error (MSE), root mean squared error (RMSE), and mean absolute error (MAE). Gradient Boosting achieved the best predictive results, with an R^2 of 0.988, MSE of 1.39×10^{-6}, RMSE of 3.72×10^{-4}, and MAE of 2.17×10^{-4}. Extreme Gradient Boosting, Random Forest, and K-Nearest Neighbours also performed well, whereas the Support Vector Regressor showed much lower predictive capability. The analysis of corrosion risk revealed that most defects fell into the low-risk category, while a significant number were classified as moderate to high risk, emphasizing the need for focused maintenance efforts. Despite the challenges posed by imbalanced datasets, the ensemble approach achieved an overall accuracy of 98.8%, showing consistent and reliable performance across all risk levels. Compared with standard maintenance practices, the proposed framework reduces unnecessary maintenance interventions, improves maintenance prioritization, and demonstrates strong potential for lowering operational costs and downtime, highlighting its effectiveness as a data-driven tool for condition-based pipeline maintenance.

1. Introduction

The global oil and gas industry relies on extensive pipeline networks to transport petroleum products efficiently. Given the vast scale and critical importance of these infrastructures, their maintenance is vital in order to ensure safe operation, prevent catastrophic failures and meet regulatory requirement [1,2]. Globally, the rate of natural gas use is rapidly increasing. In 2023, natural gas accounted for about 39 % of the United States' electric power supply; by 2024, this percentage is predicted to drop to 37 %. Its usage rate has been increasing as a result of its reduced greenhouse gas production. Pipelines are a less expensive method of transferring natural gas. But occasionally, accidents are more likely to result from this. To mention a few, the material of the pipes, their age, and their operating conditions are some of the potential reasons of this damage [3]. Damage to the pipelines may result in oil leaks that could endanger human life, have an adverse effect on company cash flows, or harm the environment [2]. In 2019 alone, pipeline incidents resulted in USD 333 million in damages and 11 fatalities. Due to pipeline leaks, the United States has seen property loss of around USD 7 billion over the last 30 years. These industries are located in desolate regions with harsh weather. As a result, people find it challenging to monitor any leaks. In order to address and identify gas leaks and avert catastrophes, numerous research have been carried out. The first leak detection system was developed by Billmann et al. using pressure and flow measurements [4]. A frequently applied technique in this area is the negative pressure wave (NPW) method, as demonstrated by

* Corresponding author.
E-mail address: adamu_22000793@utp.edu.my (A.A. Sani).

https://doi.org/10.1016/j.rineng.2026.110288
Received 22 January 2026; Received in revised form 10 March 2026; Accepted 25 March 2026
Available online 26 March 2026
2590-1230/© 2026 The Authors. Published by Elsevier B.V. This is an open access article under the CC BY-NC-ND license (http://creativecommons.org/licenses/by-nc-nd/4.0/).

Reinaldo et al. [5]. Beyond this, the literature reports several other approaches, such as time-domain reflectometry, accelerometer-based monitoring [6], distributed temperature sensing (DTS) technologies [7], passive acoustic detection systems [8], and the use of computational fluid dynamics (CFD) models [9].

In recent year, application AI [10] and machine learning (ML) techniques [11,12] has emerged as a promising solution for predicting corrosion rate and related operational parameters, thereby enabling more effective conditioned based maintenance planning in crude oil pipelines. Few research studies suggested and developed several strategies to prevent pipeline corrosion, deterioration, and failure [13,14]. To determine the maximum failure pressure and residual strength of the corroded steel pipeline, M. Ahammed developed a deterministic model decades ago [15]. In order to build prediction models for CO2 corrosion in pipelines and wells, R. Nyborg offered an overview based on a number of techniques, including mechanistic modelling and empirical correlations [16]. His group then suggested a semi-empirical method and noted challenges in forecasting corrosion in gas and oil [17]. However, a semi-empirical model was introduced in [18,19] to identify operating conditions under which water condenses either as droplets settling at the bottom of the pipe or as a continuous film along the upper wall. Subsequently, Alawi et al. ([20] developed a neural network–based approach to evaluate the effects of electromagnetic interference on gas pipeline performance. A few years later, Race, et al. [21] proposed a new scoring system aimed at estimating corrosion rates in pipelines. Based on experimental results, Papavinasam, et al. [22] formulated a model to predict internal pitting corrosion in oil and gas pipelines. In a related study, Ren, et al. [23] demonstrated that a backpropagation (BP) neural network could efficiently and rapidly predict corrosion rates in long-distance pipeline systems. At the same time, Zerpa, et al. [24] introduced an integrated model designed to forecast hydrate formation and flow restrictions in gas, oil, and water pipeline networks. Subsequently, a risk-based integrity framework was developed to support optimal replacement decisions for offshore process equipment by accounting for both the likelihood and the consequences of failures associated with time-dependent degradation. The study in [25] focuses on offshore process components that are affected by stochastic degradation mechanisms evolving over time.

However, as earlier research was primarily concerned with corrosion. Senouci et al. developed a prediction model that went beyond corrosion by incorporating additional pipeline characteristics, using both regression analysis (RA) and artificial neural networks [26,27]. The results showed that both methods performed well and achieved nearly the same level of prediction accuracy. In a subsequent study, the same authors revisited the analysis and further evaluated their findings. Based on a simplified single leak detection system, they concluded that corrosion remains one of the main contributors to pipeline failure [28,29]. To gain deeper insight into corrosion-related failures, a system-based corrosion event scenario for a subsea pipeline was developed using Bow-Tie analysis combined with Bayesian Network (BN) mapping [30]. This was followed by extensive application of Finite Element Analysis (FEA) to assess pipeline integrity, where burst pressure behaviour was investigated for various defect sizes and the results were compared with the conventional Det Norske Veritas (DNV) model [31]. Faisal and his colleagues later applied a bidirectional BN framework to evaluate the probability of pipeline failure, showing that two-way reasoning within the BN structure helped to simplify the failure probability assessment [32]. In subsequent work, they employed interpretive structural modelling (ISM), BN, and the decision-making trial and evaluation laboratory (DEMATEL) techniques to identify and analyse the key factors responsible for pipeline failures and accidents [33]. More recently, a hybrid BN–Markov approach integrated with Monte Carlo simulation was proposed for the safety assessment of offshore pipelines affected by microbiologically induced corrosion. Additionally, several researchers have introduced a range of methods addressing different challenges within oil and gas pipeline systems [34–39].

A critical gap therefore exists between high-accuracy corrosion prediction models and their practical implementation in condition-based maintenance planning. Specifically, there is insufficient integration of machine learning predictions with risk classification strategies that directly inform maintenance prioritization. Moreover, many existing studies rely on simulated or limited datasets, which may restrict real-world applicability.

To address this gap, this study proposes a data-driven ensemble machine learning framework for corrosion risk prediction and maintenance optimization in oil and gas pipelines. Unlike prior approaches that focus solely on predictive accuracy, the proposed method classifies corrosion into three actionable risk levels: low, moderate, and high. These classifications are directly linked to maintenance scheduling decisions, enabling a shift from reactive maintenance to proactive, risk-informed asset management. The central hypothesis of this study is that ensemble machine learning models trained on real operational data can achieve high predictive accuracy while simultaneously supporting structured, risk-based maintenance optimization. To test this hypothesis, twelve regression models were developed and evaluated using robust statistical performance metrics, including Mean Squared Error, Root Mean Squared Error, Mean Absolute Error, and the coefficient of determination. The predictive outputs were then integrated into a maintenance prioritization framework to assess practical applicability.

The main objectives of this research are:

1. To develop and evaluate multiple machine learning models for corrosion rate prediction using real operational pipeline data.
2. To compare ensemble and non-ensemble techniques in terms of predictive performance and robustness.
3. To classify corrosion risk into actionable categories that support condition-based maintenance scheduling.
4. To demonstrate how predictive analytics can reduce unnecessary interventions, optimize resource allocation, and lower operational downtime.

By bridging predictive modelling with maintenance decision-making, this study advances current research beyond model comparison toward a practical and implementable framework for pipeline integrity management. The proposed approach contributes to the growing body of knowledge on data-driven asset management and supports the transition from reactive to proactive maintenance strategies in critical energy infrastructure systems.

2. Machine learning in pipeline integrity management

The complexity and size of oil and gas pipeline networks make it difficult and expensive to rely on human operators to do inspections [40]. Governments, businesses, and academics have been trying to automate these checks for a long time. They want to reduce expenses and improve the quality of inspections. To address this issue, they have investigated several strategies, including machine learning, analytical modelling, and numerical calculations [41,42]. Machine learning has emerged as an important field of research and is increasingly being applied across industries, including for the inspection of oil and gas pipelines. Although probabilistic models are frequently used to anticipate pipeline failures, machine learning techniques are currently thought to be more appropriate for anticipating unforeseen problems in this difficult and expansive sector [43]. By analysing pressure data and precisely locating the leak using in-depth sound analysis, machine learning techniques were utilized to automatically identify leaks [44,45]. According to recent studies, machine learning methods have a lot of promise for automatically displaying, identifying, and classifying pipeline defects [46]. Additionally, it was noted that using machine learning to assess the safety of oil and gas pipelines is a developing field of study with a dearth of literature that addresses all of the present research issues [47]. But according to Wang and Cheng [45] traditional machine

learning models will be limited in terms of accuracy and efficiency since they rely only on data that already exists without incorporating engineering concepts. Crucial steps in creating a prediction model include choosing the right model and fine-tuning its parameters (Ramadhan et al., 2021).

2.1. Constrain of conventional maintenance methods of pipeline

Condition-based maintenance, reliability cantered, time-based maintenance are the three common methods used to keep crude oil pipelines in good condition. Every technique has distinct qualities and limitations that reduce the efficiency and cost-effectiveness of maintenance operations [48–50].

Condition-Based Maintenance (CBM) enhances productivity using sensors and regular inspections to assess pipeline problems in real time. Even though CBM has advantages, it also relies on human perception and established thresholds. This reliance may not always lead to optimal maintenance procedures since Personal decision and threshold designs may not always be sufficient for recognizing complex deterioration patterns.

Time based maintenance Regardless of the pipeline's current state, it follows a predetermined maintenance plan. This approach often results in increased operating costs, unnecessary maintenance, reducing resource usage without necessarily enhancing pipeline dependability, particularly when it provides frequent inspections and interventions.

A more sophisticated strategy called reliability-centered maintenance, or RCM2, incorporates risk-based evaluations and prioritization methods. It helps with strategic decision-making by concentrating maintenance efforts on important pipeline components.

Despite these drawbacks, ML-based predictive maintenance offers a workable alternative by dynamically enhancing maintenance schedules using historical data and real-time monitoring.

2.2. Machine learning algorisms

Supervised learning is used in this work. Discrete variable classification and regression tasks are handled by supervised learning techniques [51]. Machine learning techniques are described here.

A. Artificial Neural Network (ANN)

ANN are computational models inspired by the way the human brain is organized and functions, and they are widely used to solve complex problems encountered in real-world applications. ANNs are suitable for both regression and classification analyses. Structurally, they are composed of input, hidden, and output layers, where the neurons in the hidden layers are essential for learning relationships and accurately transforming input data into corresponding outputs [52]. The neurons compute a weighted sum of the input signals, denoted by *xi* for =1, 2, …, *I*. The sum is subsequently passed through an activation function, as outlined in Eq. (1). Here, w_i symbolizes the weight estimated, f denotes the activation function, and (α_0) represents the term bias.

$$y = f\left(\sum_{i\,is\,1}^{I} w_i x_i + \alpha_0\right) \tag{1}$$

B. Support Vector Machine (SVM)

Support Vector Machines, grounded in statistical learning theory, are supervised learning techniques commonly applied to regression problems [53]. In support vector regression (SVR), the core idea is to map the input variable x into a higher-dimensional feature space $\phi(x)$ through the use of a kernel function. This transformation, which is represented in Equation (2), enables more effective modelling of complex relationships.

$$y = \text{ sign } \left(\mathrm{w}^{\mathrm{T}}\phi\ (\mathrm{x})\ w + b\right) \tag{2}$$

The ideal bias vector(b) and the ideal weight vector(w), which are represented by the constant w, are derived by minimizing the sum of empirical risk.

C. Gradient Boosting

Gradient boosting is a robust machine learning technique applied to both classification and regression problems. It is particularly effective when strong correlations exist among the input variables [54,55]. The method builds a highly accurate predictive model by combining a series of simple learners, where each successive model is trained to address and reduce the errors made by earlier ones. Consider a sequence of classifiers, each denoted by the letters YGM and Gm and carrying a different weight. Eq. (3) describes how weighted majority voting is used to combine these classifiers to get a final, robust mode.

$$y_{Gm} = \overset{\text{argmin}}{\gamma} \sum_{x_i \in R_{Gm}} L(y_i, F_{m-1}(x_i) + \gamma) \tag{3}$$

D. Linear Regression

Linear Regression is a popular statistical technique used for making predictions. It works by analysing a dataset of input variables to forecast an output variable [55]. The advantage of this method is that it provides weights to each input variable once the model is trained. Selecting the appropriate weight function is important because it helps us determine how much each input parameter matters. Eq. (4) shows the formula for Linear Regression.

$$y = \beta_o + \beta_1 x + \varepsilon \tag{4}$$

In this case, x represents the input, y the expected output, βo the bias, and β1 the coefficient corresponding to the input x each.

E. K-Nearest Neighbor (KNN)

As stated by [56–58], the K-Nearest Neighbours (KNN) algorithm is a widely used classification technique utilized in many different domains. It is well-liked due to its traditional approach and adaptability. The technique of KNN operates on the basis that objects belonging to the same category are usually found close to one another. Finding the K closest data points by calculating distances and evaluating dataset commonalities is what makes it effective. Equation 5 is widely used for the mathematical representation of this process.

$$\widehat{y}_q = sign\left(\sum_{x_i \in N_k(x_q)} y_i\right) \tag{5}$$

The categorized example, the set of KNN x_q and y_i, represented by $N_k(x_q)$ are used to determine the projected class for observation q, which is written as $\widehat{y_q}$. The observed class for observation i, represented by yi, is usually either -1 or 1.

F. Random Forest (RF)

Machine learning technique that makes use of decision trees(DT) is Random Forest(RF), according to [55,59]. To address both classification and regression issues, it functions as an ensemble, combining many decision trees. In the process of creating each decision tree, several variables are chosen at random. Equation 6 explains how to prevent overfitting by using several trees with distinct predictors. The variables x' is represented by the decision tree $f_b(x')$. Finally, we may calculate the means result of every tree f_b related to x' to predict the results for x'. The technique can also be used to determine the significance of a feature in a dataset.

$$\widehat{f} = \frac{1}{B}\sum_{b=1}^{B} f_b(x') \tag{6}$$

G. AdaBoost (AB)

AdaBoost is an ensemble boosting technique used in machine learning [60]. By integrating multiple weaker learners, it produces stronger ones. According to Equation 7, the weak learner's output errors have an impact on the alpha parameter estimation.

$$H(x) = sign\left(\sum_{t=1}^{T} \alpha_t h_t(x)\right) \tag{7}$$

The output of the t th weak learner is denoted by $H(x)$, while α_t represents the weight assigned by AdaBoost

3. Methodology

This study's methodology includes essential steps to developing a machine learning model. Pre-processing and obtaining the required dataset are the first steps in the procedure. Training the proposed model and assessing how it performed is the next stage. The procedure is described in further detail in this section, and an outline of the study's approach is shown in Fig. 1.

3.1. Description of the data

This study used the [61], an open-source dataset from GitHub. This dataset is available to the public for machine learning and statistical research, among other academic uses. The dataset, which has eight numerical parameters and 10,293 occurrences. The dataset can be applied to both regression and classification tasks and is commonly divided into training and testing sets. Table 1 provides a summary and description of all variables. The severity of the corrosion defect in the pipeline is primarily described by the CR, an indicator of the level of corrosion. The corrosion risk level (CR) is used as the target variable in our prediction models to assess pipeline integrity and also provides

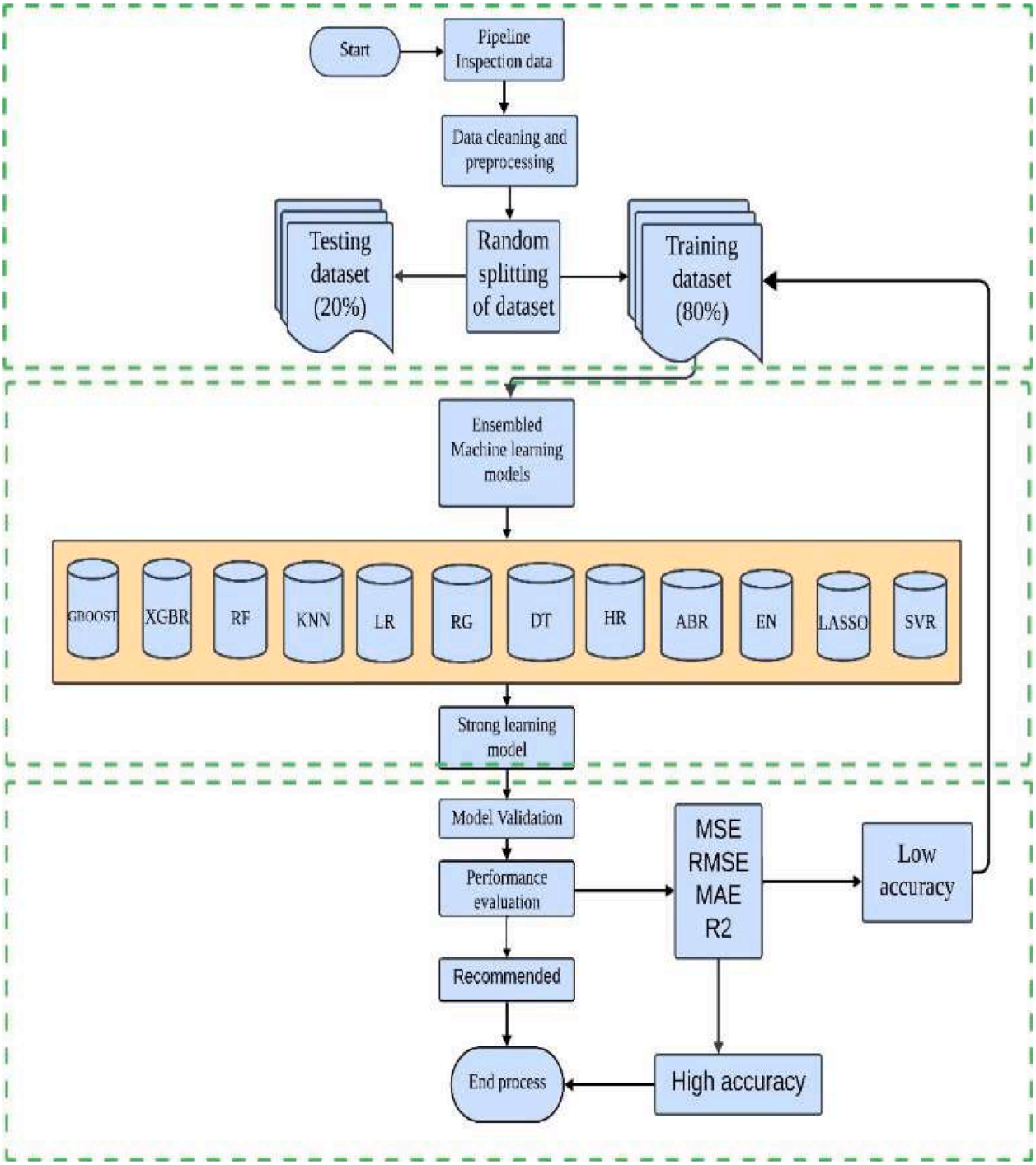


Fig. 1. Methodological Flow chart.

Table 1
List of dataset features.

Feature	Description	Type
Wellhead Temperature (°C)	The measured temperature at the wellhead.	Input
Wellhead Pressure (psi)	The operating pressure recorded at the wellhead.	
MMCFD Gas	Daily gas production rate in million standard cubic feet.	
BOPD	Daily oil production measured in barrels.	
BWPD	Daily water production measured in barrels.	
BSW	Basic sediment and water content in the produced fluids.	
CO_2 Mole Fraction	The molecular concentration of carbon dioxide.	
Gas Gravity	The specific gravity of the produced gas.	
CR	Measured corrosion defect or corrosion risk level.	Output

insights for evaluating pipeline condition and planning maintenance actions.

3.2. Feature engineering

After the data acquisition phase is completed, the next step involves Feature Engineering, wherein the data undergoes essential preprocessing. To improve the quality of the raw historical data, reduce noise, and extract important information that is essential to the prediction model's performance, this entails extracting, manipulating, and choosing applicable features. In this instance, the corrosion risk is the goal output variable for the machine learning prediction model; that is, the model will be trained to predict the risk level of corrosion by taking into account different input characteristics that were seen or recorded during inspection activities.

To predict oil pipeline maintenance schedules using corrosion risk as the output variable, a correlation between corrosion risk level and the accumulation of deposits or contaminants in the pipeline was establishes. A lower corrosion risk suggests stable pipeline operation and may extend the interval before maintenance is required, whereas a higher corrosion risk level indicates potential deterioration that could necessitate earlier maintenance intervention.

A. Data Preprocessing

In machine learning, different input variables often have varying units of measurement, which can affect the results of data analysis. To reduce the impact of these differing units, the data should be preprocessed using normalization, as shown in Eq. (8), or standardization, as presented in Eq. (9) [62].

A. Data Preparation and Cleaning

- Min-Max normalization

$$x' = (x - X_\min)/(X\max - X_\min) \tag{8}$$

- Z-score standardization

$$x' = (x - \mu)/\sigma \tag{9}$$

3.3. Model development and hyperparameter tuning

The dataset was scaled and normalized to ensure uniformity among variables for corrosion rate prediction. Corrosion assessment was formulated as a regression task, with 80 % of the data used for training and 20 % for testing. The following twelve regression-based machine learning algorithms were evaluated: Gradient Boosting, Extreme Gradient Boosting, Random Forest, K-Nearest Neighbors, Decision Tree, Linear Regression, Ridge Regression, AdaBoost, Huber Regression, Lasso Regression, Elastic Net Regression, and Support Vector Regression.

Hyperparameters were optimized using grid search with five-fold cross-validation, maximizing the coefficient of determination (R^2). Table 2 summarizes the main models and key hyperparameter settings.

Predicted corrosion rates were then mapped to three risk categories: low (<0.1 mm/year), moderate (0.1–0.5 mm/year), and high (>0.5 mm/year). Maintenance intervals were assigned accordingly: high risk required immediate action, moderate risk within 1–2 years, and low risk beyond two years. Risk categorization is implemented as a post-prediction decision framework, not as a standalone classification task.

3.3.1. Proposed approach: ensemble-based machine learning

Ensemble learning improves prediction accuracy by merging the results of multiple machine learning models, which also enhances the performance of each individual model. The word "ensemble" refers to this collection of models used to make predictions [42,63]. In AI and data science, ensemble techniques combine several simpler algorithms, such as Decision Trees, to boost the overall quality of predictions and analysis [64]. These methods are gaining popularity in many applications because they generally perform better and are more adaptable than using a single model [64]. As noted by Seghier, et al. [65], weak learners can be assembled into either sequential or parallel ensemble models, depending on the combination method. In parallel ensembles, similar weak learners are trained at the same time but independently, with their outputs typically averaged to produce the final prediction. Sequential ensembles, in contrast, use boosting techniques to reduce both bias and variance in the results [65].

3.3.2. Evaluation of the developed models' performance

Statistical adequacy used in models for maintenance schedule prediction based on corrosion rate was confirmed by the study using a variety of performance indicators, such as R^2, RMSE, MSE, and MAE. RMSE shows how closely the predicted values match the actual values by taking the square root of the average squared differences between them. Equation 10 presents the RMSE formula.

$$RMSE = \sqrt{\frac{1}{n}\sum_{i=1}^{m}\left(y_i^{abs} - y_i^{pnd}\right)^2} \tag{10}$$

RMSE and MAE are used as performance evaluation metrics. RMSE measures how closely the predicted values match the observed data by computing the square root of the mean of the squared differences between each prediction and its corresponding actual value.

Table 2
Summarizes the main models and key hyperparameter settings.

Model	Main Tuned Hyperparameters	Final Settings / Notes
Gradient Boosting	learning rate, n_estimators, max_depth	0.05, 200, 5
Extreme Gradient Boosting	learning rate, n_estimators, max_depth	0.05, 200, 5
Random Forest	n_estimators, max_depth, min_samples_split	200, 10, 2
K-Nearest Neighbors	n_neighbors, weights	5, distance
Decision Tree	max_depth, min_samples_split	default / minimal tuning
Linear Regression	–	default
Ridge Regression	alpha	default / optimized if needed
AdaBoost	n_estimators, learning_rate	100, 0.1
Huber Regression	epsilon	default
Lasso Regression	alpha	default / optimized if needed
Elastic Net Regression	alpha, l1_ratio	default / optimized if needed
Support Vector Regression	C, gamma, kernel	1, scale, rbf

y_i^{abs} = observed data, y_i^{ped} = n represents the total number of data points, and data predicted

In regression modelling, the mean squared error (MSE) measures the average deviation between predicted outcomes and observed values, thereby reflecting how accurately the model fits the data. Its mathematical expression is presented in Equation (11).

$$MSE = \frac{1}{n}\sum_{i=1}^{n}(\tilde{y}_i - y_i)^2 \tag{11}$$

Mean absolute error (MAE) is used to quantify the spread between predicted results and observed data. It is obtained by computing the mean of the absolute differences between the estimated values and the actual measurements. The corresponding formulation is presented in Eq. (12).

$$MAE = \frac{\sum_{i=1}^{n}|y_i^{ab} - y_i^{pect}|}{n} \tag{12}$$

R^2 evaluates the degree of agreement between the model's predicted results and the observed data by analysing the variability between them. Its formulation is presented in Eq. (13).

$$R^2 = 1\frac{\sum_{i=1}^{n}\left(y_i^{abs} - y_i^{pred}\right)^2}{\sum_{i=1}^{n}\left(y_i^{abs} - y_i^{obs}\right)^2} \tag{13}$$

y_i^{obs} represents the mean value of the observed dataset.

4. Result and discussion

This study employs a range of visualization techniques and evaluation metrics to evaluate the performance of machine learning algorithms in predicting corrosion conditions in oil pipelines. The analysis revealed meaningful correlations within the dataset and provided valuable insights for developing risk-based maintenance schedules based on prediction accuracy. In this study, corrosion assessment is formulated as a regression-based prediction problem, where machine learning models estimate the continuous corrosion rate. The predicted corrosion rate is subsequently categorized into corrosion risk levels (low, moderate, and high) using predefined engineering thresholds to support maintenance decision-making. Accordingly, regression evaluation metrics, including the coefficient of determination (R^2), Root Mean Square Error (RMSE), and Mean Absolute Error (MAE), are employed to assess model performance, while corrosion risk classification serves as a post-prediction decision framework rather than an independent classification modelling task.

4.1. Analysis of feature importance

Fig. 2 presents the relative importance of the features in predicting pipeline corrosion rates. The results show that wellhead temperature (°C) and wellhead pressure (psi) are the most influential variables, followed by MMCFD gas, gas gravity, and BOPD. Less influential features include BWPD, BSW, and CO_2 mol. The feature importance values quantify the contribution of each parameter to the regression model, with longer bars indicating greater influence. This analysis provides a physically interpretable insight, as wellhead temperature and pressure are known to accelerate corrosion, while flow rate and gas composition influence chemical interactions within the pipeline. Prioritizing these features enhances model understanding, guides maintenance decision-making, and supports risk-informed pipeline management.

4.2. Matrix of correlation

The heatmap provides an illustration of the connections among the variables in the dataset. Fig. 3 illustrates this. Stronger associations between variables are shown by deeper colour tones, whether the relationship is positive or negative, while pale or greyish colours indicate weaker links. Heatmaps support the easy identification of multicollinearity, reveal underlying patterns among variables, and help in selecting important features for further analysis or model development. Moreover, including the correlation coefficient values on the heatmap provides precise numerical insight, leading to clearer interpretation and more informed decisions.

4.3. Corrosion defect trends

Fig. 4a displays a line graph showing how CR–corrosion defect values change with their indices, while Fig. 4b presents the same data as a

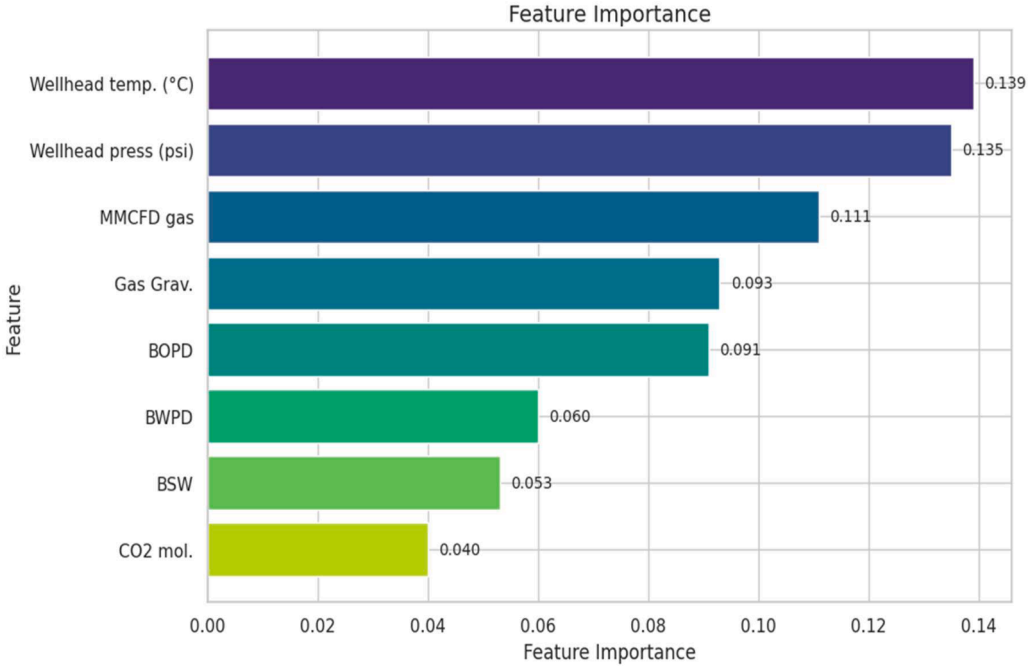


Fig. 2. Feature importance of pipeline operational parameters in corrosion risk prediction.

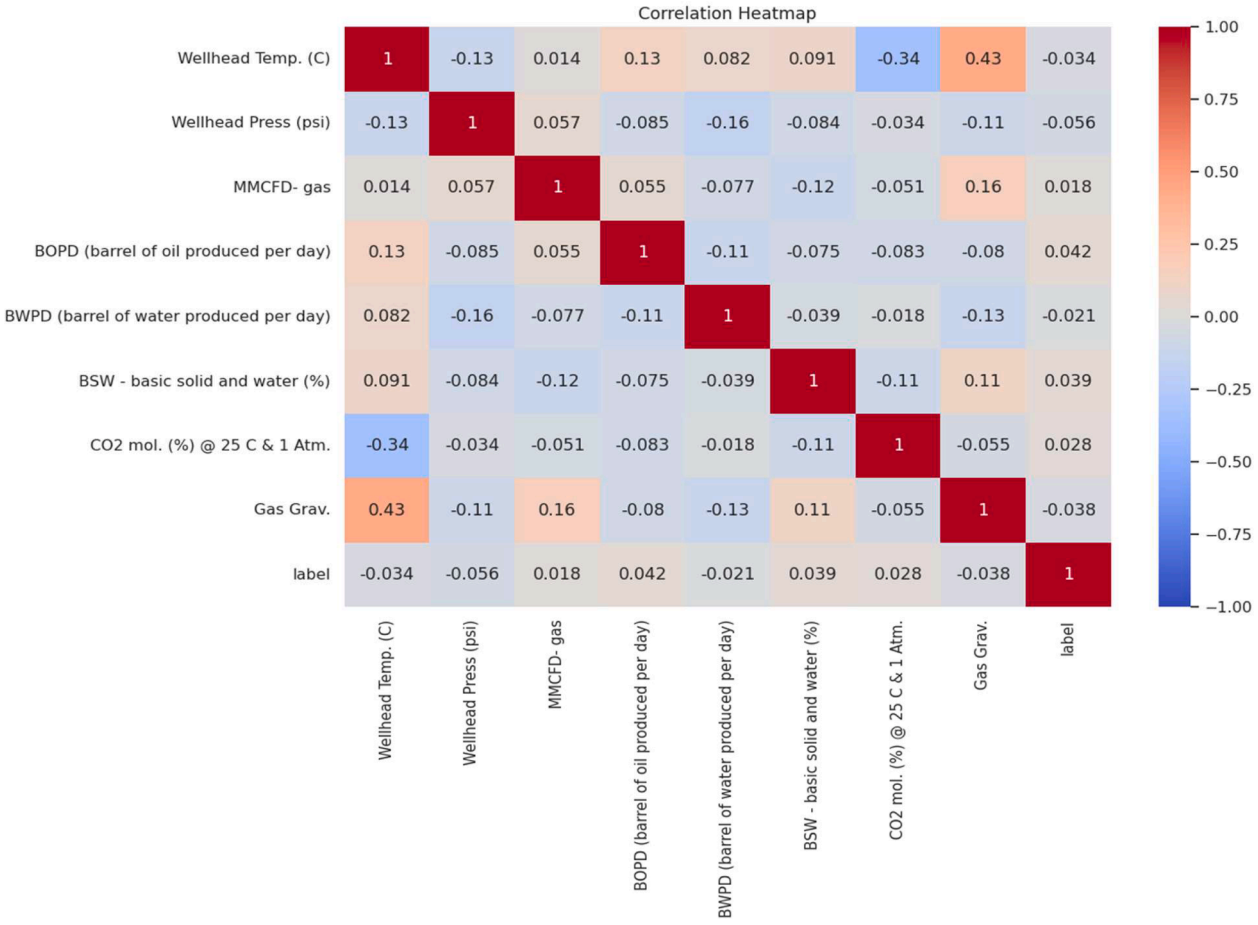


Fig. 3. Variables Correlation Matrix.

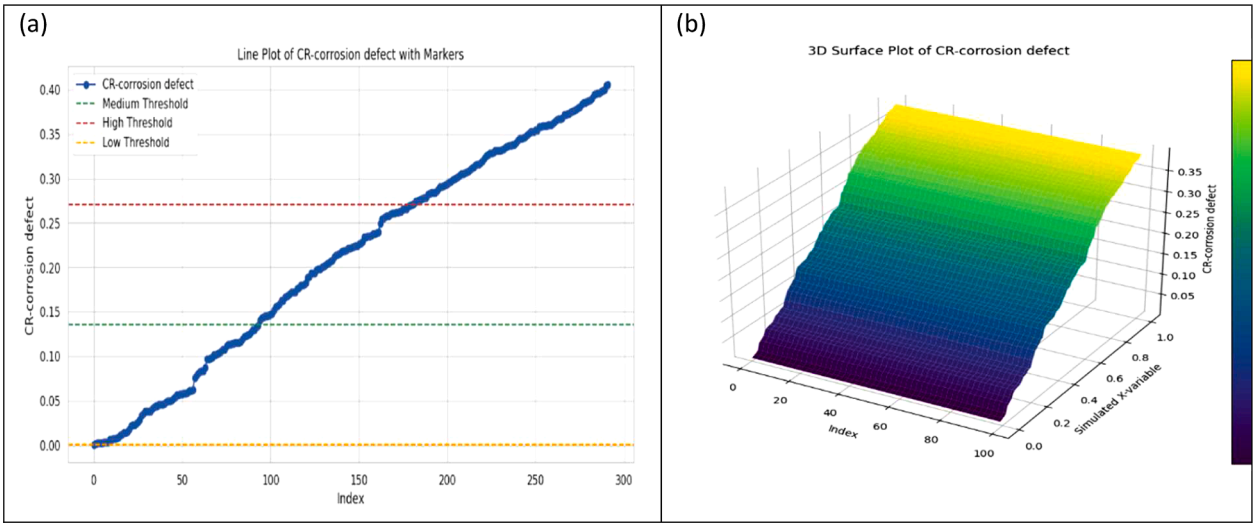


Fig. 4. Visualization of corrosion defect values. (a) Line plot showing defect variations and (b) 3D surface plot representing spatial distribution.

three-dimensional surface plot. In the line graph, three dashed horizontal lines mark the corrosion risk levels: low risk at $y = 0.0009$ (orange), medium risk at $y = 0.1357$ (green), and high risk at $y = 0.2705$ (red). The data show that most pipeline sections fall into the low-risk category, as the majority of CR–corrosion defect values remain below the medium-risk limit. Near the middle of the dataset, the trend crosses the green reference line, reflecting a more balanced spread of corrosion defect values at that point. Only a few data points approach the high-risk region, as indicated by the red line. The gradual upward trend observed in the CR-corrosion defect values suggests that timely preventive maintenance and regular inspection programs are necessary to mitigate progressive damage and sustain pipeline safety.

4.4. Corrosion defect distribution

Fig. 5a uses a histogram to show how CR-corrosion defects are distributed, while Fig. 5b provides a three-dimensional surface representation of the same data. Most of the observed defects are clustered at the lower range of values, particularly between 0.00 and 0.05, which suggests that inspection practices and preventive measures are generally effective. Three dashed reference lines mark the boundaries between low (0.02), medium (0.25), and high corrosion severity. Most of the observed defects fall below the medium range, suggesting that corrosion along the pipeline is generally low to moderate in severity. As the values increase toward the upper end, the number of occurrences drops noticeably, indicating that severe corrosion is not widespread. The overall form of the distribution suggests that corrosion behaviour is influenced by several interacting factors, which merits further detailed investigation. In addition, the distribution exhibits multiple peaks around 0.05, 0.20, and 0.35, revealing a multimodal pattern. Overall, the findings identify areas where corrosion control has been successful as well as zones where further reduction in defect frequency is still required.

4.5. Impact of an imbalance in corrosion levels on model performance and prediction accuracy

Figs. 6a and 6b illustrate the number of occurrences in the three corrosion risk categories: low, medium, and high. It can be seen that the high corrosion category has the largest number of instances, followed by the low category, while the medium category has the fewest, slightly below the high level. This shows that the dataset is unbalanced, with more frequent severe corrosion events, even though they are not necessarily critical. Such an uneven distribution may affect the model's predictions if no corrective measures, such as balancing, are applied, as the model could be biased toward the high corrosion category. When evaluating the model, this imbalance should be considered, with careful attention to precision, recall, and overall accuracy for each corrosion level. Analysts should also account for this pattern during data analysis, since it could reduce the model's ability to correctly identify medium and low corrosion cases.

4.6. Model performance and reliability using balanced corrosion label distribution

The distribution of corrosion labels across the three risk levels (0, 1, and 2) is depicted in a bar chart in Fig. 7(a–b). Each category has approximately over 100 instances, suggesting that the dataset is evenly distributed. A successful and objective machine learning model must achieve this balance. The model may favour the dominant class if the dataset is imbalanced, meaning some classes have higher occurrence than others. This would lead to minority classes having inaccurate predictions. Cross-class generalization and effective model learning are made possible by a balanced dataset. Furthermore, k-fold cross-validation ensures correct performance metrics since each fold accurately depicts the dataset's distribution. This balance ensures the model's effectiveness in practical applications by enhancing its validity and reliability.

4.7. Performance of the ensemble models' technique

Ensemble techniques improve overall performance of the model by combining predictions from multiple individual algorithms. K-Nearest Neighbor, Linear Regression, Ridge, Decision Tree, Huber Regressor, ElesticNet, Lasso, Support Vector Regressor, XGBoost Regressor, Random Forest, AdaBoost Regressor, Gradient Boosting, and Lasso.

A. Comparison of Actual and Predicted Corrosion Risk Levels

Fig. 8 illustrates the comparison between actual and predicted corrosion risk levels across different machine learning models, providing insight into each predictive accuracy of the model. With the actual corrosion risk level on the x-axis and the expected value on the y-axis, every point of data represents a pair of real and predicted values. Perfect prediction, in which the expected and actual values are precisely the same, is shown by a dashed diagonal line. The data points above the line imply overestimation, whereas those below show underestimation. The accuracy of the model is reflected in the degree of dispersal around the line; tighter clustering indicates greater prediction performance and more agreement.

B. Line Plot Comparison for Prediction Accuracy

Fig. 9 presents a line plot containing 10 individual data points, each labelled with a serial number and showing both the actual value and the predicted value. This plot visually compares the observed and predicted results, making it easy to identify where the model's predictions are accurate and where they deviate. It provides a clear assessment of the model's performance, highlighting both its strengths and weaknesses.

B. Ensemble models learning curve

Fig. 10 (a-h) illustrates the learning curve for the ensemble models' technique. The plots show the comparison between the training and cross-validation scores for each regressor with the number of training examples: Gradient Boosting, Random Forest, and K-Nearest Neighbors are all models that have had very consistent, high scores on both training

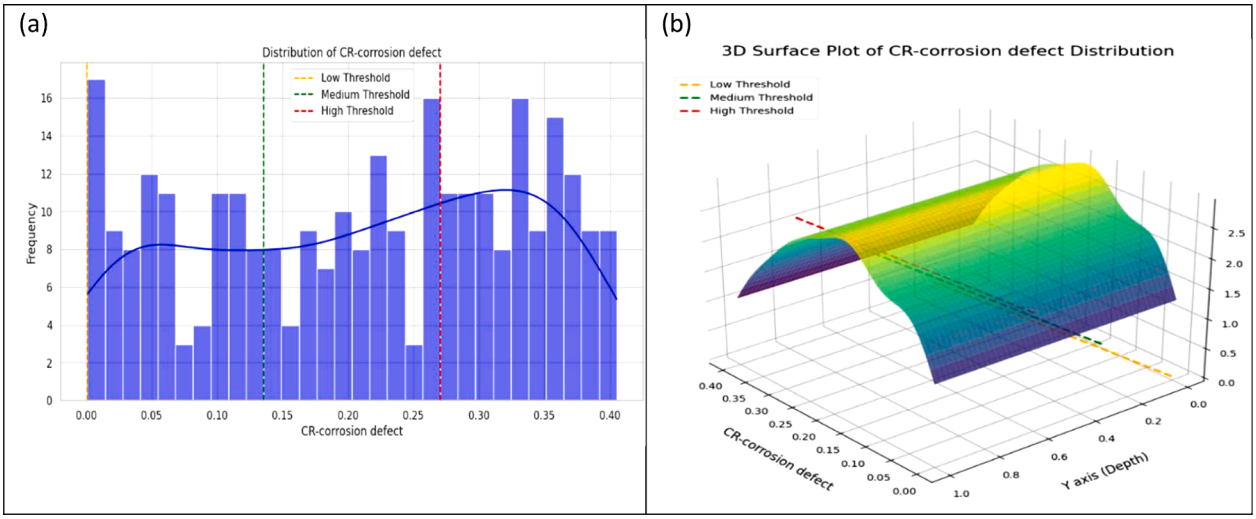


Fig. 5. Visualization of defect patterns. (a) Corrosion defect distribution and (b) 3D surface Plot.

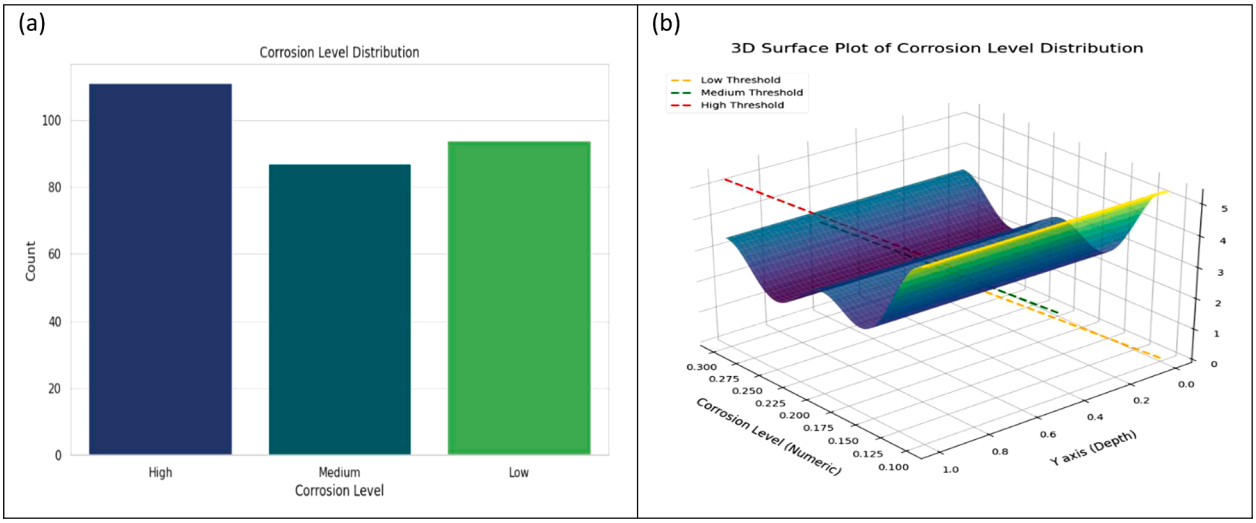


Fig. 6. Visualization of corrosion level distribution. (a) 2D distribution of corrosion levels and (b) 3D surface plot illustrating spatial variations.

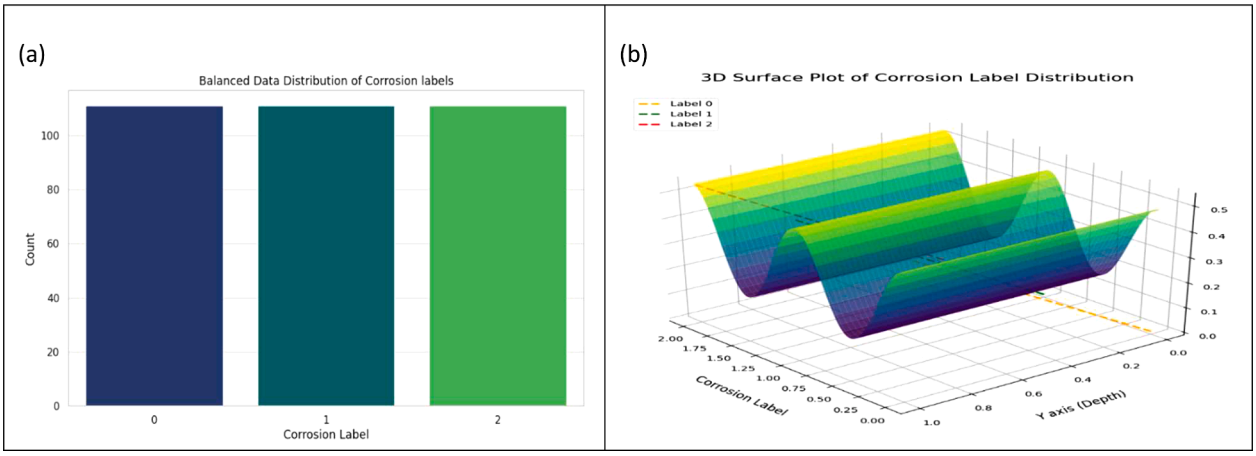


Fig. 7. Visualization of corrosion risk levels. (a) Balanced distribution of corrosion risk levels and (b) 3D surface plot illustrating spatial trends.

and validation; hence, they generalize very well to new data. Whereas Lasso and Support Vector Regressor have low validation scores, along with a higher gap between the training and the validation, hence suggesting underfitting. Moving on to the AdaBoost Regressor and Huber Regressor, they are pretty good but a little overfitting because their training score is higher compared to their validation score. These results generally point to the fact that some models learn better from data, and hence choosing an appropriate model requires due attention.

C. Comparison of Performance Metrics

Table 3 summarizes the evaluation of different models in this study, based on MAE, RMSE, and R^2, to assess their predictive performance. This performance matrix, include MSE, R^2, RMSE and MAE, are often used for performance assessment of regression models. Higher R^2 values and lower MSE, RMSE, and MAE values are often indicative of better model performance [66]. According to the evaluation's findings, the best-performing models are Gradient Boosting and XGBoost Regressor, while Support Vector Regressor performs less satisfactorily.

C. Comparison of ensemble models using RMSE, MAE, MSE, and R^2 scores.

Fig. 11 shows the performance of different ensemble models using RMSE, MAE, MSE, and R^2 as evaluation metrics. Each bar represents a model, and the length of the bar indicates how well that model predicts the outcomes, with longer bars showing stronger performance. This figure makes it easier to see which model performs best for a specific task or dataset, aiding in the selection of the most suitable machine learning model.

4.8. *Optimized maintenance schedule with ML-based predictive modelling*

The proposed machine learning framework enables condition-informed maintenance scheduling by predicting pipeline corrosion rate and supporting timely intervention decisions. Unlike conventional maintenance approaches based on fixed inspection intervals, the model allows maintenance activities to be planned according to predicted pipeline degradation trends, thereby improving operational planning and resource utilization.

(A) Improved Maintenance Prediction Accuracy

Using historical operational data, the model predicts corrosion rate with high accuracy ($R^2 = 0.988$), enabling early identification of pipeline segments requiring preventive maintenance.

(B) Adaptive Maintenance Scheduling

Predicted corrosion rates are categorized into low, moderate, and high-risk levels linked to corresponding maintenance intervals, allowing maintenance activities to be prioritized according to pipeline condition.

(C) Improved Operational Efficiency

Condition-informed maintenance planning minimizes unnecessary inspections and supports efficient allocation of maintenance resources.

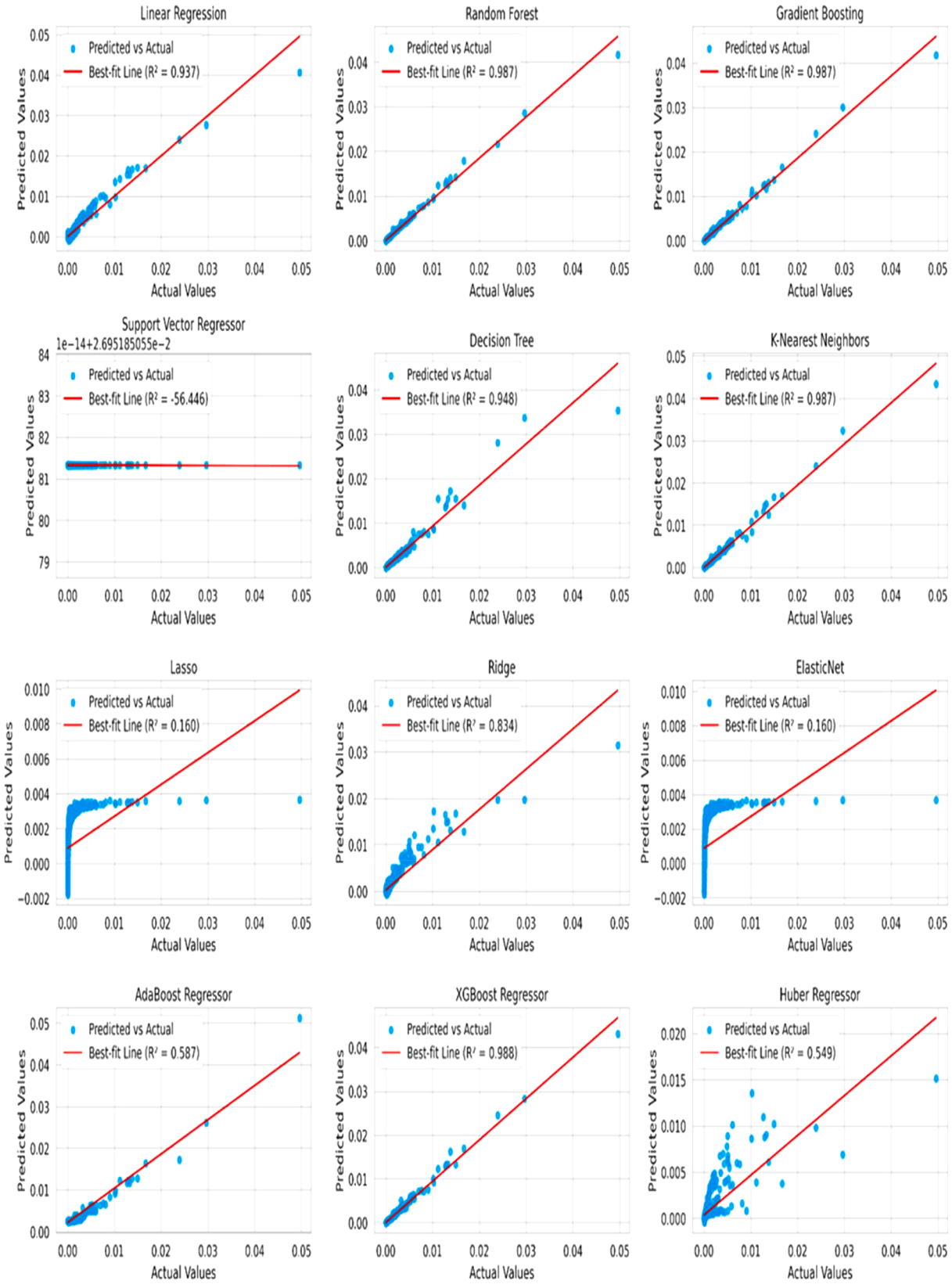


Fig. 8. Comparison of actual and predicted corrosion risk levels using various machine learning models.

4.9. Comparison of ML-based and traditional maintenance approaches

While Section 4.6 presents the operational application of the proposed predictive maintenance framework, this section provides a conceptual comparison with conventional maintenance strategies from a decision-analysis perspective.

The proposed framework differs from traditional maintenance strategies by incorporating a data-driven prediction of corrosion rate to

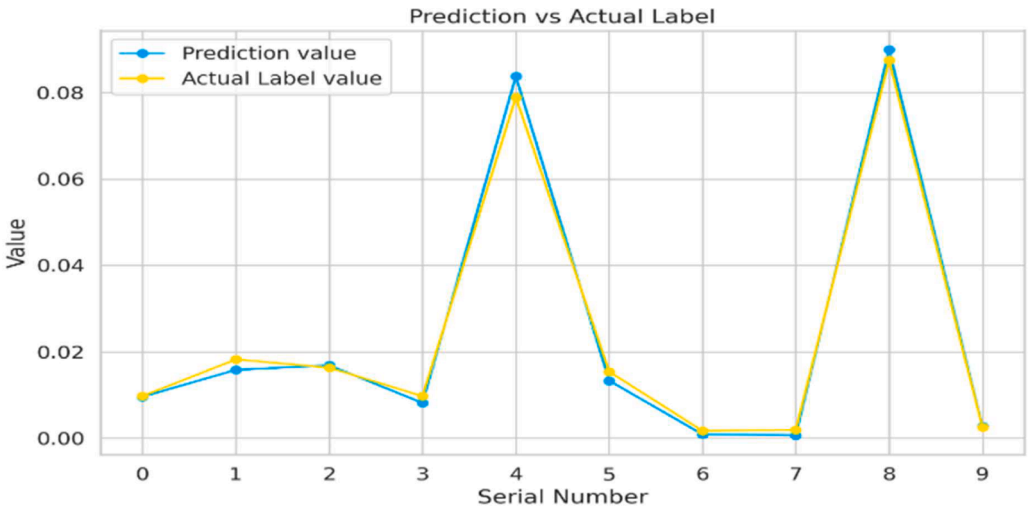


Fig. 9. Variables correlation matrix.

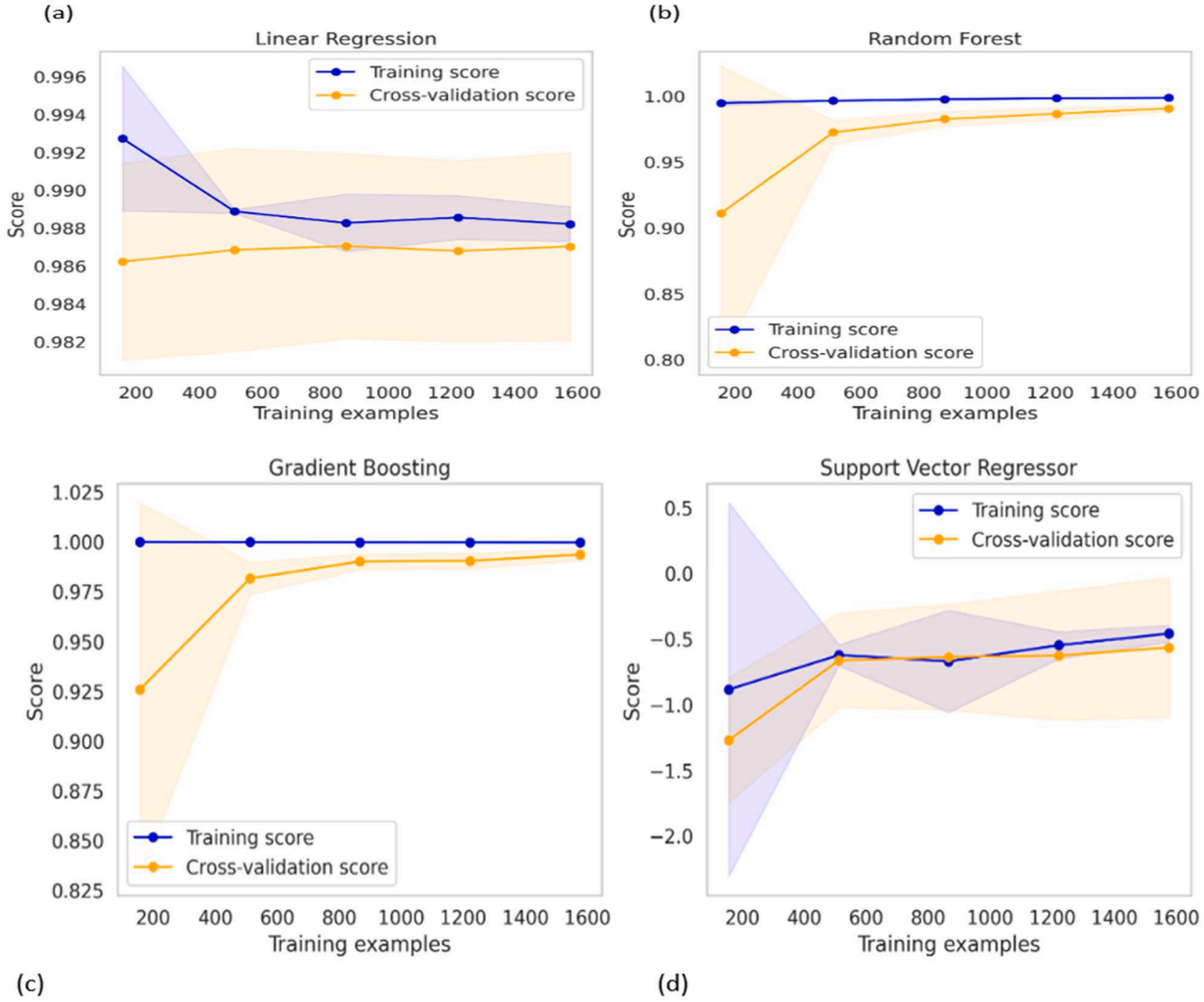


Fig. 10. (a-h): Training progression curve for the ensemble modelling approach.

support maintenance timing decisions. Unlike Time-Based Maintenance (TBM), Condition-Based Maintenance (CBM), or Reliability-Centered Maintenance (RCM2), which rely on predefined intervals or expert judgment, the present approach utilizes predicted corrosion rate as a

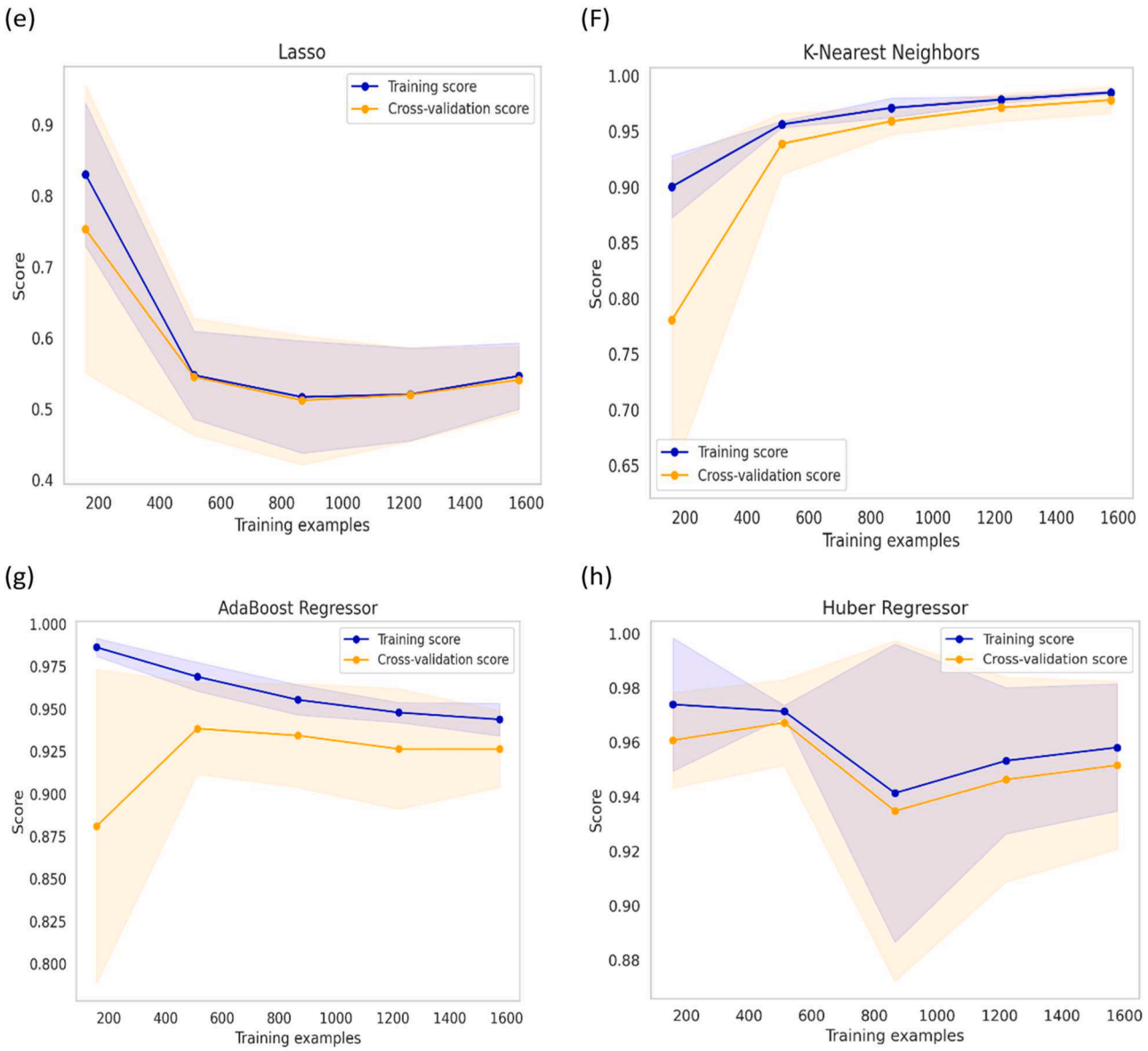


Fig. 10. (*continued*).

Table 3
Performance metrics for ensemble models.

Rank	Acronym	Model Name	MSE	RMSE	MAE	R^2
1	GB	Gradient Boosting	0.00000139	0.000372	0.000217	0.988
2	XGB	Extreme Grad. Boosting	0.00000153	0.000391	0.000221	0.987
3	RF	Random Forest	0.00000155	0.000394	0.000223	0.987
4	KNN	K-Nearest Neighbours	0.00000161	0.000401	0.000244	0.987
	DT	Decision Tree	0.00000625	0.000791	0.000454	0.948
5	LR	Linear Regression	0.00000751	0.000866	0.000647	0.937
6	RD	Ridge Regression	0.0000198	0.001409	0.001229	0.834
7	AB	AdaBoost	0.0000492	0.00222	0.001548	0.587
8	HR	Huber Regression	0.0000537	0.00292	0.002104	0.549
10	Lass	Lasso Regression	0.0001	0.00715	0.003031	0.160
11	EN	Elastic Net Regression	0.0001	0.00847	0.003075	0.160
12	SVR	Support Vector Regression	0.000685	0.02619	0.02489	-56.45

continuous engineering indicator for risk-informed scheduling.

To provide a quantitative interpretation, the maintenance decision can be conceptually expressed as:

$$C_{total} = C_m + P_f(CR) X C_f \quad (14)$$

where C_m represents preventive maintenance cost, C_f denotes failure consequence cost, and $P_f(CR)$ is the probability of failure as a function of predicted corrosion rate (CR). Improved prediction accuracy reduces uncertainty in estimating P_f, thereby supporting more rational mainte-

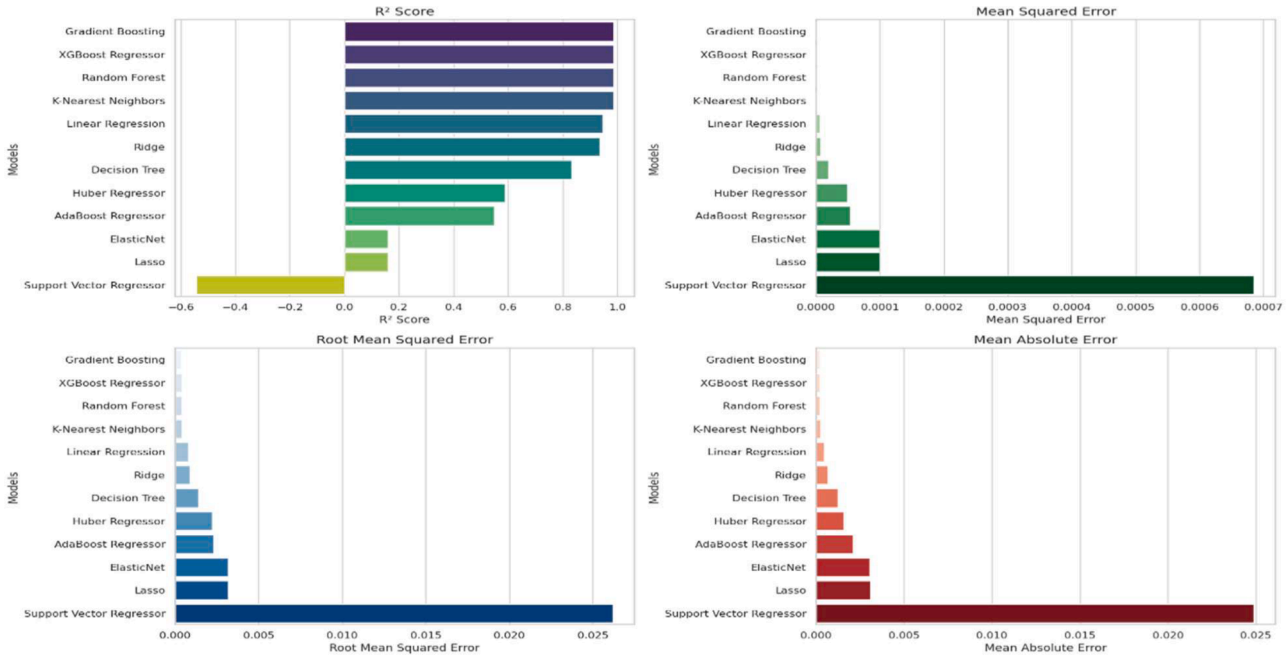


Fig. 11. Models' performance evaluation using RMSE, MAE, MSE and R^2 scores.

nance timing decisions.

It is important to note that the present study does not implement a full probabilistic life-cycle cost optimization model. Rather, it provides a predictive foundation upon which such cost–risk optimization models can be integrated in future work.

Compared with conventional maintenance strategies, the ML-driven approach offers:

- Improved estimation of degradation trends,
- Reduced reliance on fixed maintenance intervals,
- Enhanced prioritization of high-risk pipeline segments,
- A structured quantitative basis for future optimization integration.

4.9.1. *Illustrative calculation of expected maintenance cost and downtime*

To provide a quantitative illustration of the proposed framework, the total expected cost for a pipeline segment is calculated using the risk-based cost model:

$$C_{\text{total}} = C_m + P_f(\text{CR}) \times C_f \tag{15}$$

where:

- $C_{\text{total}}$is the total expected cost
- $C_m$is the planned maintenance cost
- $P_f(\text{CR})$is the probability of failure as a function of predicted corrosion rate (CR)
- $C_f$is the consequence or cost of failure

Example Calculation:
Consider a pipeline segment with the following parameters:

- Corrosion rate: 0.6 mm/year (High risk)
- Planned maintenance cost (C_m): $10,000
- Failure probability (P_f): 0.75
- Failure consequence (C_f): $50,000

$$C_{\text{total}} = 10,000 + 0.75 \times 50,000 = 47,500 \tag{16}$$

This example shows that the expected cost for a high-risk segment is $47,500. Applying this calculation across all pipeline segments allows for risk-informed maintenance prioritization, reducing unnecessary interventions and downtime. Table 4 shows the Illustrative Segments.

4.10. *Limitations and future work*

While the proposed framework demonstrates strong predictive performance for corrosion rate estimation, several limitations should be acknowledged. First, model validation was conducted using train–test split and cross-validation; external field validation on independent pipeline systems is recommended for future studies. Second, the maintenance framework functions as a regression-driven decision-support tool rather than a full probabilistic cost–risk optimization model. Future research will integrate life-cycle cost analysis, failure probability modelling, and uncertainty quantification to develop a comprehensive reliability-based maintenance optimization framework.

Table 4
Conceptual comparison of maintenance strategies.

Maintenance Strategy	Decision Basis	Maintenance Frequency	Quantitative Prediction Capability	Expected Operational Impact*
TBM	Fixed time interval	High (4–6/year)	None	Risk of unnecessary interventions
CBM	Sensor thresholds	Moderate (2–4/year)	Limited	Improved responsiveness
RCM2	Risk assessment + expert judgment	Moderate (1–3/year)	Semi-quantitative	Better prioritization
ML-Based Predictive (This Study)	Predicted corrosion rate (regression-based)	Adaptive (condition-dependent)	High ($R^2 \approx$ 0.99 for best model)	Potential improvement in maintenance prioritization and downtime reduction

5. Conclusion

This study explored the use of ML algorithms to predict corrosion risk levels in crude oil pipelines, aiming to support data-driven and risk-informed maintenance planning. By classifying corrosion rates into low, moderate, and high-risk levels, the developed models enable more accurate scheduling of maintenance activities, reducing unnecessary interventions while ensuring timely response to pipeline deterioration. Twelve regression models were developed and assessed using key performance metrics, including Mean Squared Error (MSE), Root Mean Squared Error (RMSE), Mean Absolute Error (MAE), and the coefficient of determination (R^2). Among these models, the Gradient Boosting demonstrated the best predictive performance, achieving an R^2 of 0.988, MSE of 1.39×10^{-6}, RMSE of 3.72×10^{-6} and MAE of 2.17×10^{-6}. Extreme Gradient Boosting, Random Forest, and K-Nearest Neighbours also exhibited strong performance, each attaining an R^2 value of 0.988. In contrast, Support Vector Regressor had the weakest performance with a negative R squared value of -56.446 and significantly higher error metrics, indicating poor predictive capability. The analysis further showed that approximately 64 percent of the pipeline segments were classified under the moderate corrosion risk category, requiring maintenance within one to two years. Compared with conventional maintenance practices, the proposed framework reduces unnecessary maintenance interventions, improves maintenance prioritization, and demonstrates strong potential for lowering operational costs and downtime, highlighting its effectiveness as a data-driven tool for condition-based pipeline maintenance. The findings also emphasize the value of using accurate predictive models, particularly ensemble learning methods, in optimizing pipeline maintenance decisions. This approach supports a shift from reactive to proactive maintenance strategies in the oil and gas sector. Further research involving field validation is recommended to further enhance the robustness and practical implementation of the proposed framework. Compared with conventional maintenance practices, the proposed framework reduces unnecessary interventions and improves maintenance prioritization, leading to lower operational costs and downtime, while underscoring the role of accurate ensemble-based predictive models in supporting effective condition-based pipeline maintenance.

Conflict of interest statement

The authors declared no potential conflict of interest with respect to the research authorship and or publication of this article.

Funding

This research is supported by YUTP grant [YUTP-FRG 1/2022 (015LCO-421)], Universiti Teknologi PETRONAS Malaysia.

Data Availability

Data is available upon request from the corresponding author.

CRediT authorship contribution statement

Adamu Abubakar Sani: Writing – original draft, Methodology, Formal analysis, Conceptualization. **Mohamed Mubarak Abdul Wahab:** Writing – review & editing, Supervision. **Nasir Shafiq:** Writing – review & editing. **Kamaluddeen Usman Danyaro:** Writing – review & editing. **Yusuf Aliyu:** Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgment

The authors wish to express their sincere thanks to the Management of Universiti Teknologi PETRONAS, Malaysia, for their support in carrying out this research. They are also grateful to the reviewers for their insightful comments, which helped improve the overall quality of the study.

Data availability

Data will be made available on request.

References

[1] H. Ku, et al., Flow characteristics of gas and liquid in pipeline revealed by machine learning on distributed acoustic sensing data, Geoenergy Sci. Eng. 245 (2025) 213518, https://doi.org/10.1016/j.geoen.2024.213518.
[2] G. Wang, Q. Cheng, W. Zhao, Q. Liao, H. Zhang, Review on the transport capacity management of oil and gas pipeline network: challenges and opportunities of future pipeline transport, Energy Strateg. Rev. 43 (2022) 100933, https://doi.org/10.1016/j.esr.2022.100933.
[3] A.A. Sani, et al., A multi-level classification model for corrosion defects in oil and gas pipelines using meta-learner ensemble (MLE) techniques, J. Pipeline Sci. Eng. 5 (2) (2025) 100244, https://doi.org/10.1016/j.jpse.2024.100244.
[4] L. Billmann, R. Isermann, Leak detection methods for pipelines, Automatica 23 (3) (1987) 381–385.
[5] R.A. Silva, C.M. Buiatti, S.L. Cruz, J.A. Pereira, Pressure wave behaviour and leak detection in pipelines, Comput. Chem. Eng. 20 (1996) S491–S496, https://doi.org/10.1016/0098-1354(96)00091-9.
[6] H. Jin, L. Zhang, W. Liang, Q. Ding, Integrated leakage detection and localization model for gas pipelines based on the acoustic wave method, J. Loss. Prev. Process. Ind. 27 (2014) 74–88, https://doi.org/10.1016/j.jlp.2013.11.006.
[7] L. Wang, S.C. Narasimman, S.R. Ravula, A. Ukil, Water ingress detection in low-pressure gas pipelines using distributed temperature sensing system, IEEe Sens. J. 17 (10) (2017) 3165–3173, https://doi.org/10.1109/JSEN.2017.2686982.
[8] Y. Mahmutoglu, K. Turk, A passive acoustic based system to locate leak hole in underwater natural gas pipelines, Digit. Signal. Process. 76 (2018) 59–65, https://doi.org/10.1016/j.dsp.2018.02.007.
[9] L. Xinhong, C. Guoming, Z. Renren, Z. Hongwei, F. Jianmin, Simulation and assessment of underwater gas release and dispersion from subsea gas pipelines leak, Process Saf. Environ. Protect. 119 (2018) 46–57, https://doi.org/10.1016/j.psep.2018.07.015.
[10] S.M. Ali, A.U. Rahman, G. Kabir, S.K. Paul, Artificial intelligence approach to predict supply chain performance: implications for sustainability, Sustainability. 16 (6) (2024) 2373, https://doi.org/10.3390/su16062373.
[11] M.M. Khan, I. Bashar, G.M. Minhaj, A.I. Wasi, N.U.I. Hossain, Resilient and sustainable supplier selection: an integration of SCOR 4.0 and machine learning approach, Sustain. Resilient. Infrastruct. 8 (5) (2023) 453–469, https://doi.org/10.1080/23789689.2023.2165782.
[12] A. Shehadeh, O. Alshboul, Enhancing engineering and architectural design through virtual reality and machine learning integration, Buildings 15 (3) (2025) 328, https://doi.org/10.3390/buildings15030328.
[13] J.C. Velázquez, F. Caleyo, A. Valor, J.M. Hallen, Predictive model for pitting corrosion in buried oil and gas pipelines, Corrosion 65 (5) (2009) 332–342, https://doi.org/10.5006/1.3319138.
[14] L. Li, W. Gong, J. Li, Prediction on service life of concrete pipeline buried in chlorinated environment under nonuniformly distributed earth pressure, Constr. Build. Mater. 243 (2020) 118162, https://doi.org/10.1016/j.conbuildmat.2020.118162.
[15] M. Ahammed, Prediction of remaining strength of corroded pressurised pipelines, Int. J. Press. Vessels Pip. 71 (3) (1997) 213–217, https://doi.org/10.1016/S0308-0161(96)00081-6.
[16] R. Nyborg, Overview of CO2 corrosion models for wells and pipelines. NACE Corrosion, NACE, 2002 pp. NACE-02233.
[17] R. Nyborg and A. Dugstad, "Reliability and limitations of corrosion prediction tools for oil and gas pipelines", 2004.
[18] F. Vitse, K. Alam, Y.M. Gunaltun, D. Larrey, Semi-empirical model for prediction of the top of the line corrosion risk, NACE Corrosion (2002), https://doi.org/10.5006/C2002-02245 pp. NACE-02245.
[19] A. Shehadeh, O. Alshboul, Game theory integration in construction management: a comprehensive approach to cost, risk, and coordination under uncertainty, J. Constr. Eng. Manage 151 (5) (2025) 04025039, https://doi.org/10.1061/JCEMD4.COENG-15109.
[20] S. Al-Alawi, A. Al-Badi, K. Ellithy, An artificial neural network model for predicting gas pipeline induced voltage caused by power lines under fault conditions, COMPEL-Int. J. Comput. Math. Electr. Electron. Eng. 24 (1) (2005) 69–80, https://doi.org/10.1108/03321640510571057.
[21] J.M. Race, S. Dawson, L. Stanley, S. Kariyawasam, Predicting corrosion rates for onshore oil and gas pipelines, Int. Pipeline Conf. 42622 (2006) 385–396, https://doi.org/10.1115/IPC2006-10261.

[22] S. Papavinasam, A. Doiron, R.W. Revie, Model to predict internal pitting corrosion of oil and gas pipelines, Corrosion 66 (3) (2010), https://doi.org/10.5006/1.3360912, 035006-035006-11.
[23] C.-y. Ren, W. Qiao, X. Tian, Natural gas pipeline corrosion rate prediction model based on BP neural network. Fuzzy Engineering and Operations Research, Springer, 2012, pp. 449–455, https://doi.org/10.1007/978-3-642-28592-9_47.
[24] L.E. Zerpa et al., "Predicting hydrate blockages in oil, gas and water-dominated systems", Offshore Technology Conference, 2012: OTC, pp. OTC-23490-MS, doi: https://doi.org/10.4043/23490-MS.
[25] P. Thodi, F. Khan, M. Haddara, Risk based integrity modeling of offshore process components suffering stochastic degradation, J. Qual. Maint. Eng. 19 (2) (2013) 157–180, https://doi.org/10.1108/13552511311315968.
[26] A. Senouci, M. Elabbasy, E. Elwakil, B. Abdrabou, T. Zayed, A model for predicting failure of oil pipelines, Struct. Infrastruct. Eng. 10 (3) (2014) 375–387, https://doi.org/10.1080/15732479.2012.756918.
[27] A. Shehadeh, O. Alshboul, M.M. Taamneh, A.Q. Jaradat, A.H. Alomari, Enhanced clash detection in building information modeling: leveraging modified extreme gradient boosting for predictive analytics, Results. Eng. 24 (2024) 103439, https://doi.org/10.1016/j.rineng.2024.103439.
[28] A. Aljaroudi, F. Khan, A. Akinturk, M. Haddara, P. Thodi, Risk assessment of offshore crude oil pipeline failure, J. Loss. Prev. Process. Ind. 37 (2015) 101–109, https://doi.org/10.1016/j.jlp.2015.07.004.
[29] O. Alshboul, A. Shehadeh, G. Almasabha, Reliability of information-theoretic displacement detection and risk classification for enhanced slope stability and safety at highway construction sites, Reliab. Eng. Syst. Saf. 256 (2025) 110813, https://doi.org/10.1016/j.ress.2025.110813.
[30] Y. Yang, F. Khan, P. Thodi, R. Abbassi, Corrosion induced failure analysis of subsea pipelines, Reliab. Eng. Syst. Saf. 159 (2017) 214–222, https://doi.org/10.1016/j.ress.2016.11.014.
[31] H. Liu, F. Khan, P. Thodi, Revised burst model for pipeline integrity assessment, Eng. Fail. Anal. 80 (2017) 24–38.
[32] W. Wang, K. Shen, B. Wang, C. Dong, F. Khan, Q. Wang, Failure probability analysis of the urban buried gas pipelines using bayesian networks, Process. Saf. Environ. Protect. 111 (2017) 678–686, https://doi.org/10.1016/j.psep.2017.08.040.
[33] F. Li, W. Wang, S. Dubljevic, F. Khan, J. Xu, J. Yi, Analysis on accident-causing factors of urban buried gas pipeline network by combining DEMATEL, ISM and BN methods, J. Loss. Prev. Process. Ind. 61 (2019) 49–57.
[34] M. Mahmoodian, C.Q. Li, Failure assessment and safe life prediction of corroded oil and gas pipelines, J. Petrol. Sci. Eng. 151 (2017) 434–438, https://doi.org/10.1016/j.petrol.2016.12.029.
[35] A. Mohamed, M.S. Hamdi, S. Tahar, A hybrid intelligent approach for metal-loss defect depth prediction in oil and gas pipelines, in: Intelligent Systems and Applications: Extended and Selected Results from the SAI Intelligent Systems Conference (IntelliSys) 2015, Springer, 2016, pp. 1–18, https://doi.org/10.1007/978-3-319-33386-1_1.
[36] Y.-H. Tsai, J. Wang, W.-T. Chien, C.-Y. Wei, X. Wang, S.-H. Hsieh, A BIM-based approach for predicting corrosion under insulation, Autom. Constr. 107 (2019) 102923, https://doi.org/10.1016/j.autcon.2019.102923 vol.
[37] M.E.A.B. Seghier, B. Keshtegar, K.F. Tee, T. Zayed, R. Abbassi, N.T. Trung, Prediction of maximum pitting corrosion depth in oil and gas pipelines, Eng. Fail. Anal. 112 (2020) 104505, https://doi.org/10.1016/j.engfailanal.2020.104505.
[38] M. Salemi, H. Wang, Fatigue life prediction of pipeline with equivalent initial flaw size using bayesian inference method, J. Infrastruct. Preserv. Resil. 1 (2020) 1–15, https://doi.org/10.1186/s43065-020-00005-y.
[39] W. Xiang, W. Zhou, A nonparametric bayesian network model for predicting corrosion depth on buried pipelines, Corrosion 76 (3) (2020) 235–247, https://doi.org/10.5006/3421.
[40] A.S. Abubakar, M.M.B.A. Wahab, N. Shafiq, K.U. Danyaro, A.U. Adebanjo, Integrating life cycle cost analysis into pipeline asset Integrity management: a comprehensive approach in decision support systems, J. Hunan Univ. Nat. Sci. 51 (1) (2024), https://doi.org/10.55463/issn.1674-2974.51.1.10.
[41] A.A. Sani, et al., A critical and bibliometric review of life cycle cost analysis integration into decision support systems for Pipeline Asset Integrity management, J. Pipeline Sci. Eng. (2025) 100307, https://doi.org/10.1016/j.jpse.2025.100307.
[42] A. Shehadeh, O. Alshboul, Enhancing occupational safety in construction: predictive analytics using advanced ensemble machine learning algorithms, Eng. Appl. Artif. Intell. 159 (2025) 111761, https://doi.org/10.1016/j.engappai.2025.111761.
[43] A.A. Soomro, A.A. Mokhtar, J.C. Kurnia, N. Lashari, H. Lu, C. Sambo, Integrity assessment of corroded oil and gas pipelines using machine learning: a systematic review, Eng. Fail. Anal. 131 (2022) 105810, https://doi.org/10.1016/j.engfailanal.2021.105810.
[44] R.P. da Cruz, F.V. da Silva, A.M.F. Fileti, Machine learning and acoustic method applied to leak detection and location in low-pressure gas pipelines, Clean. Technol. Environ. Policy. 22 (2020) 627–638.
[45] M. Wang, J.C. Cheng, A unified convolutional neural network integrated with conditional random field for pipe defect segmentation, Comput.-Aided Civ. Infrastruct. Eng. 35 (2) (2020) 162–177, https://doi.org/10.1111/mice.12481.
[46] Y. Liu, Y. Bao, Review on automated condition assessment of pipelines with machine learning, Adv. Eng. Inform. 53 (2022) 101687, https://doi.org/10.1016/j.aei.2022.101687.
[47] A.A. Sani, M.M.A. Wahab, N. Shafiq, Z. Nizamani, W. Rafiq, A. Ullah, Evaluation of machine learning models for predicting maintenance strategies in oil and gas pipelines based on life-cycle cost analysis, J. Adv. Res. Des. 124 (1) (2025) 63–76, https://doi.org/10.37934/ard.124.1.6376.
[48] F. Basheer, M.S. Nazmudeen, and F. Mohiddin, "Comparative analysis between conventional method versus machine learning method for pipeline condition prediction", Futuristic Trends in Intelligent Manufacturing: Optimization and Intelligence in Manufacturing, pp. 61–90, 2021, doi: https://doi.org/10.1007/978-3-030-70009-6_6.
[49] J. Kim, Y. Ahn, H. Yeo, A comparative study of time-based maintenance and condition-based maintenance for optimal choice of maintenance policy, Struct. Infrastruct. Eng. 12 (12) (2016) 1525–1536, https://doi.org/10.1080/15732479.2016.1149871.
[50] A.A. Sani, M.M.A. Wahab, N. Shafiq, Comparative analysis of machine learning algorithms for flow rate prediction in optimizing pipeline maintenance strategies, Eng. Proc. 87 (1) (2025) 37, https://doi.org/10.3390/engproc2025087037.
[51] D. Ghosh, S. Saha, Identifying river bank erosion potential zones through geo-spatial and binary logistic regression modeling approach: a case study of river Ganga in Malda district (India), Model. Earth. Syst. Environ. 10 (1) (2024) 81–98, https://doi.org/10.1007/s40808-023-01740-3.
[52] M. Romano, Z. Kapelan, and D. Savić, "Real-time leak detection in water distribution systems", in Water distribution systems analysis 2010, 2010, pp. 1074–1082.
[53] H. Hashim, P. Ryan, E. Clifford, A statistically based fault detection and diagnosis approach for non-residential building water distribution systems, Adv. Eng. Inform. 46 (2020) 101187, https://doi.org/10.1016/j.aei.2020.101187.
[54] P. Kumari, D. Toshniwal, Extreme gradient boosting and deep neural network based ensemble learning approach to forecast hourly solar irradiance, J. Clean. Prod. 279 (2021) 123285, https://doi.org/10.1016/j.jclepro.2020.123285.
[55] O. Akinsete, A. Oshingbesan, Leak detection in natural gas pipelines using intelligent models, in: SPE Nigeria Annual International Conference and Exhibition, SPE, 2019, https://doi.org/10.2118/198738-MS. D023S009R001.
[56] T. Ravichandran, K. Gavahi, K. Ponnambalam, V. Burtea, S.J. Mousavi, Ensemble-based machine learning approach for improved leak detection in water mains, J. Hydroinformatics 23 (2) (2021) 307–323, https://doi.org/10.2166/hydro.2021.093.
[57] M. Kammoun, A. Kammoun, M. Abid, Experiments based comparative evaluations of machine learning techniques for leak detection in water distribution systems, Water Supply 22 (1) (2022) 628–642, https://doi.org/10.2166/ws.2021.248.
[58] Z. Fereidooni, H. Tahayori, A. Bahadori-Jahromi, A hybrid model-based method for leak detection in large scale water distribution networks, J. Ambient. Intell. Humaniz. Comput. 12 (2021) 1613–1629.
[59] L. Aymon, et al., Leak detection using random forest and pressure simulation, in: 2019 6th Swiss Conference on Data Science (SDS), IEEE, 2019, pp. 109–110, https://doi.org/10.1109/SDS.2019.00008.
[60] A.A. Sani, M.M.A. Wahab, S.B. Abubakar, K.U. Danyaro, N. Usman, S.A. Bustani, Machine learning-based prediction of the compressive strength of sustainable geopolymer concrete with metakaolin and fly ash using ensemble techniques, Innov. Infrastruct. Solut. 11 (1) (2026) 40, https://doi.org/10.1007/s41062-025-02430-y.
[61] https://github.com/Abimbola-ai/Oil-and-gas-pipeline-leakage. Abimbola-ai/Oil-and-gas pipeline leakage, 2020.
[62] S. Raschka and V. Mirjalili, "Python machine learning Packt Publishing Ltd", 2015.
[63] I.D. Mienye, Y. Sun, A survey of ensemble learning: concepts, algorithms, applications, and prospects, IEEe Access. 10 (2022) 99129–99149, https://doi.org/10.1109/ACCESS.2022.3207287.
[64] M.O. Edeh, et al., Artificial intelligence-based ensemble learning model for prediction of hepatitis C disease, Front. Public Health 10 (2022) 892371, https://doi.org/10.3389/fpubh.2022.892371.
[65] M.E.A.B. Seghier, D. Höche, M. Zheludkevich, Prediction of the internal corrosion rate for oil and gas pipeline: implementation of ensemble learning techniques, J. Nat. Gas. Sci. Eng. 99 (2022) 104425, https://doi.org/10.1016/j.jngse.2022.104425.
[66] Ü. Ağbulut, A.E. Gürel, Y. Biçen, Prediction of daily global solar radiation using different machine learning algorithms: evaluation and comparison, Renew. Sustain. Energy Rev. 135 (2021) 110114, https://doi.org/10.1016/j.rser.2020.110114.